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Towards the development of a deep learning based security solution for electric vehicle charging

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Abstract

Electric Vehicle Charging Station (EVCS) security is an emerging concern in our interconnected world today owing to the growing complexity and rate of cyber threats. Traditional Intrusion Detection Systems (IDS) for EV charging infrastructure have challenges in detecting novel or unexpected attacks due to their limitation of predetermined signatures as well as other limited detection capabilities. The aim of this study is towards the development of a deep learning based security solution for electric vehicle charging which detects anomalies. The Host Events subdirectory of the CICEVSE2024 dataset for electric vehicle charging infrastructure security was adopted for developing the solution. After preprocessing, the distribution of the data was optimized by applying the Standard Scaling, Log Transform and Normalization methods. Tuning of hyperparameters such as number of epochs as and batch size were carried out to develop three distinct DL models for specific target variables of State, Scenario and Label. The Deep Learning solution was thereafter tested on the bases of performance metrics such as Accuracy, F1-Score, Precision, Recall as well as ROC-AUC values. The system proved to be of excellent performance across the three target variables. Comparative analysis of the various model performances showed that the Label target variable produced the most optimal results of 0.9960, 0.9957, 0.9946 and 0.9967 respectively for accuracy, F1-score, precision and recall respectively.

Keywords

Deep Learning, State, Scenario, Label, CICEVSE2024.

1. Introduction

Electric Vehicle Charging Stations (EVCS) are on the increase due to the prevalence of electric vehicle adoption worldwide. Many countries are therefore accelerating EVCS implementation projects as swiftly as possible (Anli *et al.*, 2024; ElKashlan *et al.*, 2023; Suriya and Vijay, 2022). EVCSs serve as critical interfaces between Smart Grids (SGs) and EVs, by combining several essential components. A typical EVCS combines different key components which include; EVSE (Electric Vehicle Supply Equipment), one or more CSs (Charging Stations), a CSMS (Charging Station Management System) and a DSO (Distribution System Operator) to facilitate vehicle charging. These new charging stations leverage the Internet of Things (IoT) technology to make life more convenient and empower EVCS operators. The EVCS in this regard is an IoT device that is frequently online to improve customer service, hence, it is susceptible to cyber-physical attacks. The attacks harm the electrical grid infrastructure and

users in addition to the EVCSs. Since there are security concerns, EVCSs require novel methods for detecting attacks. Traditional intrusion detection systems (IDSs) are rule-based or signature-based, hence cannot detect attacks occurring in real-time. This limitation can make EVCS infrastructure susceptible to recent adversarial attacks being crafted by attackers. IDS that are based on anomalous behaviour detection are therefore required and such systems could be realized by leveraging Deep learning. This work is aimed towards the development of a deep learning based intrusion detection system that flags anomalies in electric vehicle charging stations. This study is important due to the advantage it would introduce by leveraging the benefits of deep learning for EVCS cybersecurity. Deep learning models can perform deep layer abstraction of EVCS data traffic and hence flag down anomalies or attacks existing within the overall data traffic in the EVCS infrastructure.

2. Literature Review

Various researchers have developed machine learning as well as deep learning based intrusion detection systems for EVCS. Makhmudov *et al.* (2025) designed a novel intrusion detection system (IDS) for EV charging infrastructures using an adaptive variant of the Random Forest classifier via application of the Adaptive Windowing drift detection technique. In multiclassification scenario, the model attained 98.40 percent accuracy, recall and precision values respectively, with F1-score of 98.31 percent. Basnet *et al.* (2021) originated a novel DL-based ransomware intrusion detection model for EVCS. All three algorithms considered, achieved around 97 percent average accuracy. Their work however, focused on ransomware threats only. Paul and Selvan (2024) developed a novel hybrid DL framework called Squirrel Search-optimized Attention-Deep Recurrent Neural Network (SS-ADRNN) for optimizing EV charging stations management. Results achieved were 99.23% accuracy and F1-score of 99.11%. They employed the CICEV2023, as one of the experimental datasets. However, this dataset is limited to DoS (Denial of Service) or Distributed DoS (DDoS) attack detection. Basnet and Ali (2020) employed Long Short Term Memory (LSTM) and Deep Neural Network (DNN) algorithms to flag down DoS attacks with respect to EVCS domain. Despite the high accuracy levels of the models, the study was limited to DoS attacks only. Furthermore, the CICIDS 2018 dataset employed by the authors is a generic IoT dataset that is not specific to EVCS systems. Benfarhat *et al.* (2025) proposed an innovative approach for diverse cyber intrusions into EVCS. For the 17-class problem (all 16 attack types in the CICEVSE2024 dataset and the benign class), the model demonstrated an improvement over baseline models, with an overall accuracy of 93.9%, recall of 93.3%, precision of 94.1% and F1-Score of 93.9%.

The studies addressed above leave some room for improvement in model detection ability on one hand. Furthermore, the study by Basnet and Ali (2020) was based on a generalized IoT dataset and not an EVCS specific dataset, limiting its focus. The studies by Basnet *et al.* (2021) and Paul and Selvan (2024) considered only a single form of attack, placing a limit on their viability in a multi-attack scenario. This present study seeks to address the abovementioned gaps by developing a Deep Learning model with increased detection

accuracy on the CICEVSE2024 dataset, which contains multiple types of attack and is EVCS specific.

3. Methodology

The DNN model development was carried out in Google Colaboratory, with the entire system being developed on a single Jupyter notebook via Python 3 programming language.

3.1 Import Data and Libraries

The CICEVSE2024 dataset employed in this work was retrieved from the Canadian Institute for Cybersecurity (CIC) official website at <https://www.unb.ca/cic/datasets/evse-dataset-2024.html>. Furthermore, the following libraries were imported, numpy, pandas, StandardScaler from sklearn, Kfold, MinMaxScaler, keras from Tensorflow and matplotlib.

3.2 Data Exploration and Cleaning

The State, Scenario and Label targets present in the CICEVSE2024 dataset, were considered in this study. After feature compilation according to data classes, the dataset was filtered and null values were dropped. The number of data samples then dropped from the initial 8474 to 6107. Numerical mapping beginning from '0' was also carried out in the target columns.

3.3 Data Preprocessing

Further data preprocessing was carried out by dropping highly correlated features as well as removing data outliers via Isolation Forest method. Light Gradient Boosting Machine (LGBM) was used for feature selection in order to produce a refined feature set for the data.

In order to optimize the data distribution, Standard Scaling of data, Log Transform and Normalization techniques were employed on the data.

3.4 Data Splitting

The train_test function from sklearn was applied double-fold to realize the following ratio; 80:16:4, for training, validation and testing respectively.

3.5 Model Building

Keras was employed to build flexible and robust Sequential models by manipulating the developed tensors. A Sequential model was developed for the three CICEVSE2024 dataset target variables considered by specifying relevant parameters.

For hyperparameter tuning, the number of epochs was set to 20, the batch size was set to 64 and Adam optimizer was implemented. The hidden layers were activated with the Rectified Linear activation (ReLU) function. The output layers were thus activated with the Softmax activation and Sigmoid activation respectively (depending on multi-class or binary class outputs respectively). The loss function employed was the CategoricalCrossentropy function (GeeksforGeeks, 2025).

3.6 Model Testing

The developed models were evaluated for training and validation performances using accuracy and loss metrics respectively. Validation helps to address overfitting before testing

of the developed models. Final evaluation was then carried out on the test set via metrics such as Accuracy, F1-Score, Precision and Recall.

4. Results

4.1 Test Results for State Target Model

The test results for the developed State target model are described in Table 1. Furthermore, the model had a Receiver Operating Curve value of 0.9149.

Table 1: Test Results for State Target Model

Class	Accuracy	F1-score	Precision	Recall
0	0.9020	0.9070	1.0000	0.8298
1	0.9020	0.8966	0.8125	1.000

4.2 Test Results for Scenario Target Model

The test results for the developed Scenario target model are described in Table 2.

Table 2: Test Results for Scenario Target Model

Class	Accuracy	F1-score	Precision	Recall
0	0.9878	1.0000	1.0000	1.0000
1	0.9878	1.0000	1.0000	1.0000
2	0.9878	0.9589	0.9211	1.0000
3	0.9878	0.9663	1.0000	0.9348

4.3 Test Results for Label Target Model

The test results for the developed Label target model are described in Table 3. Furthermore, the developed model obtained an ROC curve value of 0.9967.

Table 3: Test Results for Label Target Model

Class	Accuracy	F1-score	Precision	Recall
0	0.9959	0.9946	0.9893	1.0000
1	0.9959	0.9967	1.0000	0.9935

4.4 Average Test Results for Models

The average of all test results for the developed target models is described in Table 4.

Table 4: Average Test Results for Models

Metric	State	Scenario	Label
F1-Score	0.9018	0.9813	0.9957
Accuracy	0.9020	0.9878	0.9960
Precision	0.9063	0.9803	0.9946
Recall	0.9149	0.9837	0.9967

The results in Table 4 were benchmarked against the results from a baseline Logistic Regression model as presented in Table 5 for the three targets considered.

Table 5: Baseline Logistic Regression Model Results

Metric	State	Scenario	Label
F1-Score	0.9341	0.9081	0.9056
Accuracy	0.9347	0.8951	0.9069
Precision	0.9369	0.9367	0.9152
Recall	0.9347	0.8951	0.9069

5. Discussion

5.1 Discussion of State Target Model Results

As seen in Table 1, for the charging state (0), accuracy score was 0.9020, F1-Score was 0.9070, precision was 1.0000, while recall was 0.8298 respectively. For the idle state (1), accuracy score was 0.9020, F1-score was 0.8966, precision was 0.8125, while recall was 1.0000 respectively. These test scores define an excellent model for detecting attacks at both charging and idle states of the EVCS infrastructure. Furthermore, the ROC Curve value of 0.9147 indicates that the model performs excellently well at generalization. For the metric scores considered, a closer value to 1, is a better indication of model performance.

5.2 Discussion of Scenario Target Model Results

As seen in Table 2, for Benign (0) class, the model attained an accuracy score of 0.9878 and 1.000 for F1-score, precision and recall respectively. For Cryptojacking (1) class an accuracy score of 0.9878 was also obtained as well as 1.000 for F1-score, precision and recall respectively. For the DoS (2) class, 0.9878, 0.9589, 0.9211 and 1.0000 were the scores obtained for accuracy, F1-score, precision and recall respectively. For the Recon (3) class, 0.9878, 0.9663, 1.0000 and 0.9348 were the scores obtained for accuracy, F1-score, precision and recall respectively.

The test scores obtained for this model underscore an excellent model for detecting scenario based attacks against the EVCS infrastructure, since the values obtained were very close to 1.

5.3 Discussion of Label Target Model Results

As seen in Table 3, for Benign class (0), the accuracy, F1-score, precision and recall values were 0.9959, 0.9946, 0.9893 and 1.0000 respectively. For the attack class (1), the accuracy, F1-score, precision and recall values were 0.9959, 0.9967, 1.0000 and 0.9935 respectively. The ROC Curve value of 0.9967 defines an excellent model. For the metric scores considered, a closer value to 1, is a better indication of model performance. Hence, the test scores obtained for this model underscore an excellent model for detecting label based attacks against the EVCS infrastructure.

5.4 Discussion of Average Test Results for Models

As shown in Table 4 for the average of all test results, the Label target had the best of all average performance scores, with 0.9960, 0.9957, 0.9946 and 0.9967 for accuracy, F1-score, precision and recall respectively. This indicates the Label target as the optimal target when developing IDS models for protecting EVCS infrastructure via the CICEVSE2024 dataset. Furthermore, when the results in Table 4 were compared against the baseline Logistic Regression model results in Table 5, all the developed DL models outperformed Logistic Regression for the various targets considered, except for the State target, where the ML model performed slightly better than the developed State Deep learning model. This further validates the results obtained for the deep learning solution developed in this study.

6. Conclusion and Recommendations

The aim of this study is towards the development of a deep learning based security solution for electric vehicle charging. The Host Events subdirectory of the CICEVSE2024 dataset for electric vehicle charging infrastructure security was adopted for developing the solution. After preprocessing, the distribution of the data was optimized by applying the Standard Scaling, Log Transform and Normalization methods. Tuning of hyperparameters such as number of epochs as and batch size were carried out to develop three distinct DL models for specific target variables of State, Scenario and Label. The Deep Learning solution was thereafter tested on the bases of performance metrics such as Accuracy, F1-Score, Precision, Recall as well as ROC-AUC values. The system proved to be of excellent performance across the three target variables. Comparative analysis of the various model performances showed that the Label target variable produced the most optimal results of 0.9960, 0.9957, 0.9946 and 0.9967 respectively for accuracy, F1-score, precision and recall respectively. A future direction of this study could thus be integration of the Label target model with any relevant web framework or application programming interface for deployment as a user friendly application for detecting EVCS attacks in the production environment.

Conflict of Interest

There were no conflicts of interest among the authors.

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